

Sparse Koopman Autoencoders Identify Local Dynamical Regimes in Multibasin Systems

Aidan Li, Uday Kiran Reddy Tadipatri, Mahan Fathi, Sarath Chandar, Ross Goroshin



Mila



Roadmap

1. **Motivation:** linearizing nonlinear dynamics
2. **Problem:** why multibasin systems break it
3. **Idea:** sparse supports as regime variables
4. **Model:** SKAE and its training
5. **Experiments:** forecasting, interventions, basins
6. **Discussion & outlook**

Motivation: Koopman learning

Objective

Embed nonlinear dynamics into coordinates where the evolution is **linear**, or at least locally linear.

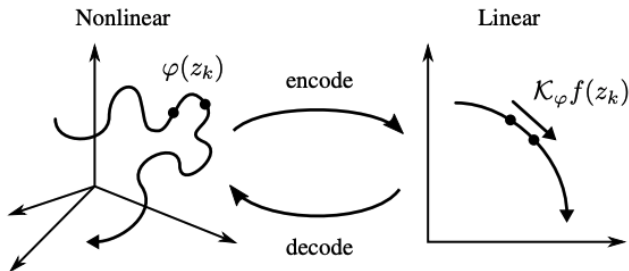


Figure from Azencot et al. (2020).

Motivation for linear coordinates

- Linear models make prediction, estimation, and control easier
- A suitable change of coordinates can substantially simplify analysis

Existence results for linearizing coordinates

- Local linearization near hyperbolic fixed points (Hartman–Grobman)
- Koopman eigenfunctions linearize on a basin of attraction...

but only *existentially*.

Good linear coordinates can exist, but how to build them from finite data?

A data-driven approach

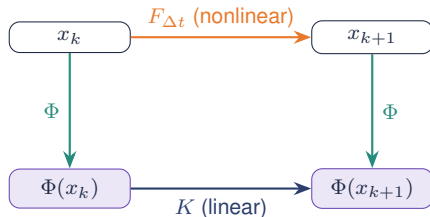
Rather than hand-designing observables,
learn the embedding directly from trajectory data.

DMD $\rightarrow$ eDMD $\rightarrow$ **Koopman autoencoders**

Koopman operator theory

Lift the nonlinear map to a **linear** operator on *observables* $g : \mathcal{X} \rightarrow \mathbb{R}$:

$$(\mathcal{K}g)(x) = g(F_{\Delta t}(x))$$



Finite approximation: find Φ (lift) and K with $\Phi(F_{\Delta t}(x)) \approx K \Phi(x)$, keeping enough to reconstruct x .

Koopman autoencoders (KAEs)

$$\hat{x}_{k+h} = D_{\phi}(K^h E_{\theta}(x_k))$$

Learn encoder E_{θ} , decoder D_{ϕ} , and one latent transition K jointly, from trajectory windows.

A key modeling constraint

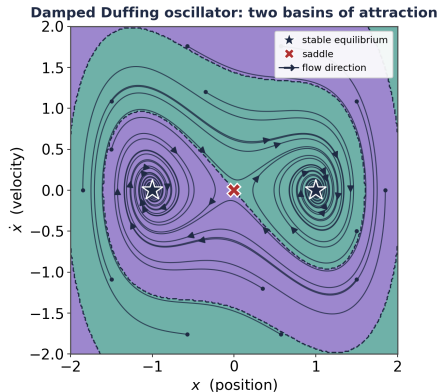
The same transition matrix K is applied at *every* rollout step.

State-dependent structure required for prediction must therefore reside in the learned representation z_k , not in a supplied regime label.

Problem: multibasin systems

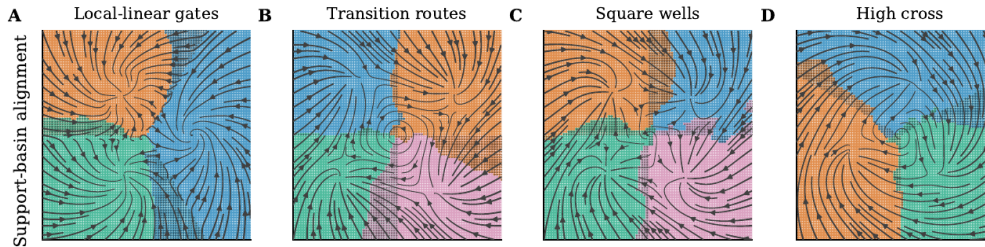
Many systems are multistable

Different initial conditions settle into **different** long-run behaviors.



Two wells, two shaded basins: where a trajectory *starts* decides which attractor it spirals into.

Coexisting attractors partition state space



Procedurally generated 2-D multibasin flows. Gray streamlines = true field; colors = distinct basins.

An impossibility result

Multibasin systems *cannot generally* admit a single finite-dimensional *global* linearization with a continuous encoder–decoder pair.

(Established in theory; confirmed empirically.)

Reframing the objective

Rather than forcing all basins through a *single* global linearization,

we seek distinct *local* linearizations.

Idea: sparse supports as regime variables

Two quantities the representation must encode

(i) **Which** local linearization is active

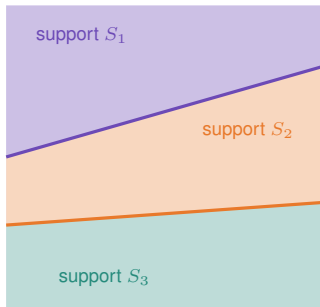
a *discrete* regime label

(ii) **Coordinates** that linearize dynamics within it

a *continuous* local chart

A sparse code encodes both quantities

the **support** (active coordinates) $\Rightarrow$ *which* regime
the **nonzero values** $\Rightarrow$ local Koopman coordinates



Different regions activate different latent supports: a union-of-subspaces structure.

Sparse coding

Given an overcomplete dictionary D , find a code with **few nonzeros**:

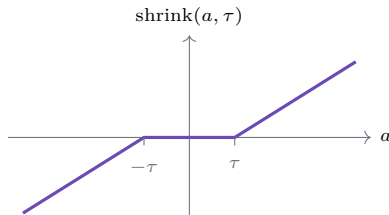
$$z^*(x) = \arg \min_z \frac{1}{2} \|x - Dz\|_2^2 + \lambda_{\text{sc}} \|z\|_1$$

The ℓ_1 penalty drives most coordinates to *exactly* zero.

Learned ISTA (LISTA): sparse coding as an encoder

Unroll the iterative solver into a fixed-depth network: $z = E_{\theta}(x)$

$$u^{(0)} = \text{shrink}(c, \tau), \quad u^{(q+1)} = \text{shrink}(S u^{(q)} + c, \tau)$$



Soft-thresholding zeros out a dead zone $[-\tau, \tau]$, creating the sparse support.

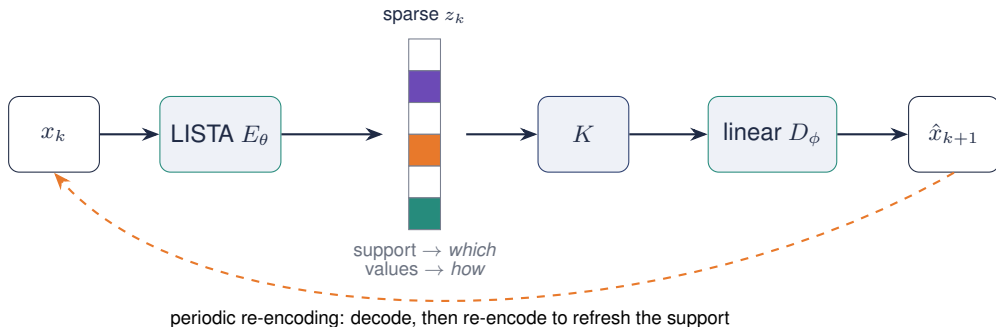
The relevant inductive bias

Thresholding makes *exact zeros*.

The encoder **explicitly selects** active coordinates,
and *learns the selection rule* from the prediction objective.

The SKAE model & training

Sparse Koopman Autoencoder (SKAE)



A LISTA encoder produces a sparse code; a unit-norm linear dictionary decodes; one matrix K advances the latent.

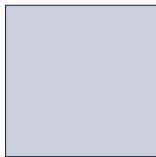
Two complementary objectives

- **Koopman:** values on each support evolve *linearly* under K
- **Sparsity:** different regimes use *different* active coordinates

Jointly, the discrete support selects the regime while the continuous values parameterize its local linear dynamics.

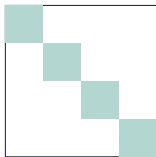
Three structures for the transition K

Dense



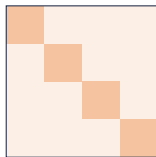
all coordinates mix

Block-diagonal



M groups, no cross-talk

Soft-block



off-block entries penalized

Varied *independently* of the encoder, to probe how block structure interacts with learned supports.

Encoders compared

- **LISTA:** sparse *by construction* (thresholding gives exact zeros)
- **Sparse MLP:** induced sparsity (ReLU with an ℓ_1 penalty)
- **Dense MLP:** no sparsity mechanism (control baseline)

Training objective: four terms

- **Prediction:** decoded rollout matches future states
- **Reconstruction:** code stays tied to the observed state
- **Latent linearity:** rolled-out latent matches encoded latent
- **Sparsity:** few active coordinates

$$\mathcal{L} = \lambda_{\text{pred}} \bar{\mathcal{L}}_{\text{pred}} + \lambda_{\text{rec}} \bar{\mathcal{L}}_{\text{rec}} + \lambda_{\text{lin}} \bar{\mathcal{L}}_{\text{lin}} + \lambda_{\text{sp}} \bar{\mathcal{L}}_{\text{sp}} + \mathcal{L}_{\text{struct}}$$

No basin labels enter training.

Support objects (1): the exact support

Computed *after* training. A Boolean active-coordinate mask:

$$m_{\text{abs},i}(x) = \mathbf{1}\{|z_i| > 10^{-3}\}, \quad S_{\text{abs}} = \{i : m_i = 1\}$$

Used to test whether the active coordinates are functionally *used*.

Support objects (2): support families

Group nearby exact supports by *Jaccard overlap* $J(m, r) \geq 0.5$.

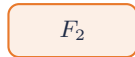
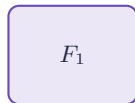
Jaccard overlap is the *intersection over union* of the two active-coordinate sets:

$$J(m, r) = \frac{|\text{active}(m) \cap \text{active}(r)|}{|\text{active}(m) \cup \text{active}(r)|}$$

exact supports



$J \geq 0.5$



Robust, basin-scale objects, with less fragmentation than raw exact supports.

Two evaluation diagnostics

- $H(B \mid F_{\text{abs}})$: uncertainty about the basin given the family

lower is better

- $|F_{\text{abs}}|$: number of families

compare to the number of basins

Basin labels are used *only* here, after training, never to train or select the model.

Periodic re-encoding refreshes the support

Every m steps the predicted latent is decoded and re-encoded, so the encoder re-selects the active support as the trajectory advances.

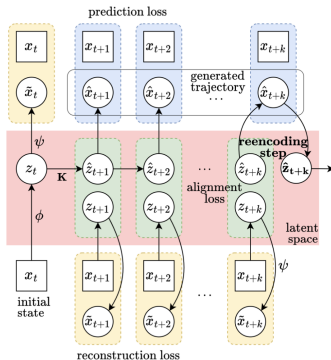


Figure from Fathi et al., ICLR 2024. Squares = ground truth, circles = model-inferred; ϕ, ψ are the encoder/decoder (our E_θ, D_ϕ).

Experiments

Benchmarks: 25 systems

- **15 procedural 2-D multibasin systems**

Gaussian wells + rotational drift; *known* basin geometry (eval only)

- **10 chaotic Dysts flows** ($dt \times 30$)

Chua, Shimizu–Morioka, and more: a long-horizon stress test

Six models compared

- **Sparse LISTA encoders**

LISTA · LISTA-BD (block-diag. K) · LISTA-SB (soft-block K)

- **Sparse MLP encoders**

ReLU + ℓ_1

Sparse MLP · Sparse MLP-BD (block-diag. K)

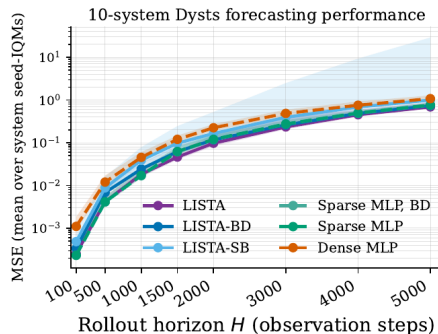
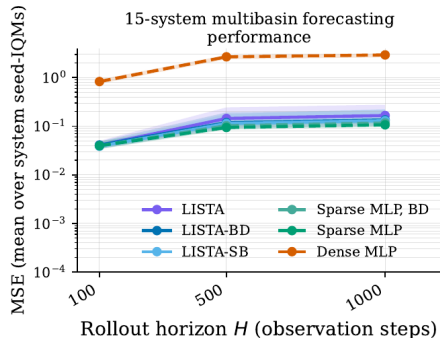
- **Dense MLP**

no sparsity: the control baseline

Experimental questions

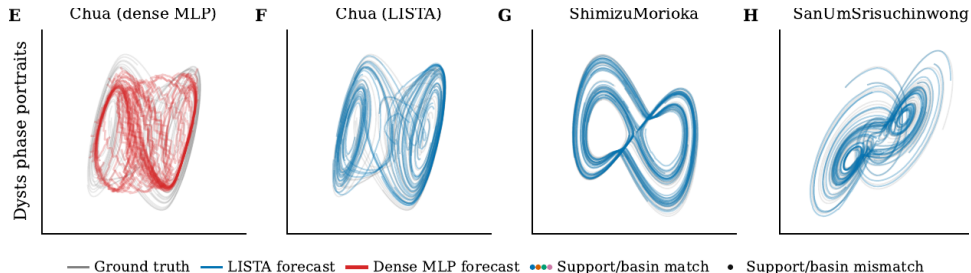
1. Do sparse encoders *forecast* better?
2. Are the supports *functionally necessary*?
3. Do support families *align with basins*?

Result 1: sparse encoders forecast better



Held-out MSE, all horizons, both benchmarks. Sparse models improve on the dense baseline at every horizon; at $H = 1000$ the dense baseline incurs **17–27 $\times$** higher error.

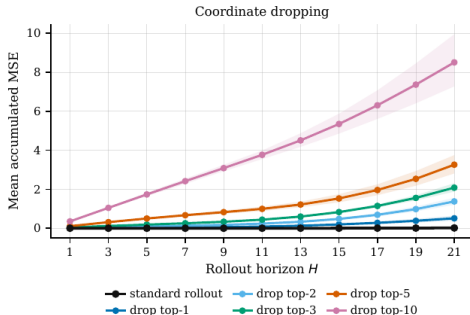
Result 1: qualitative behavior on chaotic flows



The **dense MLP (red)** loses the Chua attractor; the **LISTA (blue)** forecast stays on the true manifold. **Sparsity in the encoding is necessary for accurate long-horizon forecasts.**

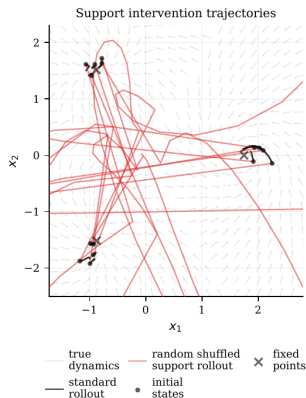
Result 2: are the supports functionally necessary?

Corrupt only the *initial active set*, keep the values fixed, then forecast.



Zeroing the top- k active coefficients sharply increases error even at short horizons; random reassignment to inactive coordinates is substantially more damaging.

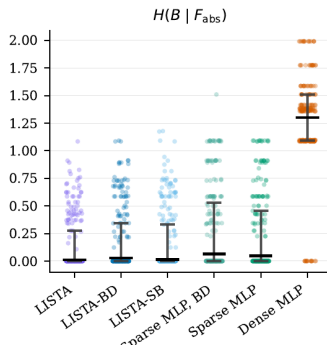
Result 2: trajectory-level effect



Randomly reassigning the active coordinates (red) drives trajectories away from the true dynamics; standard rollouts (black) remain accurate. **The active coordinates are functionally necessary, not merely correlated with the basin.**

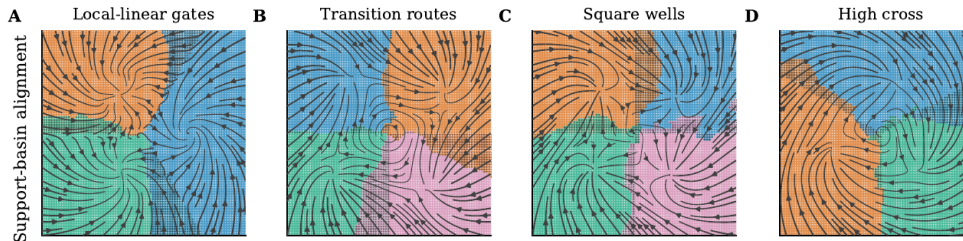
Result 3: support families identify the basin

On states *deep inside basins*, how much does a family reveal about the withheld label?
 $H(B \mid F_{\text{abs}})$, lower is better.



LISTA reaches the lowest conditional entropy (0.130); **dense MLP** sits near maximal uncertainty.

Result 3: unsupervised basin recovery



LISTA families (mapped post hoc to their dominant basin) partition the flows; colored = support/basin *matches*. LISTA exposes ≈ 4.0 families vs. the benchmark mean/median 4.20/4; **dense MLP** collapses to 1. **No basin labels are used during training.**

Discussion & outlook

A practical reconciliation

Theory: no single global finite Koopman embedding for multibasin systems.

Practice: a single sparse model *can* hold many local Koopman regimes, indexed by latent support.

Interpretability from the code structure

Sparse supports are an *interface* between continuous Koopman coordinates and discrete regimes:
a simple object you can inspect after training.

Limitations

- Interpretability shown where basin geometry is *known*
- Near separatrices, supports fragment; read them as basin-*interior* regime variables
- Deterministic benchmarks: no noise, partial observability, or control inputs yet

Future work: toward structured world models

World models (Dreamer, V-JEPA 2, Genie) predict in *latent space*, but their latent dynamics are largely *unstructured*.

- **Supports as regime latents:** rollouts that switch local linear laws *and report which law is active*; a label-free, amortized switching system
- **Control:** action-conditioned SKAE with per-regime linear MPC / LQR in the lifted space
- **Robustness & scale:** noise, partial observability, pixels; uncertainty-aware Koopman operators under distribution shift

Hafner et al. 2023–25 (Dreamer) · Assran et al. 2025 (V-JEPA 2) · Rozwood et al. 2024 (Koopman-assisted RL) · Linderman et al. 2017 (SLDS)

Summary of contributions

- (i) **Forecasting:** SKAEs outperform dense-latent KAEs at all horizons (up to $17-27\times$)
- (ii) **Necessity:** corrupting the active set sharply degrades forecasts
- (iii) **Interpretability:** support families recover withheld basin labels

Conclusion

Interpretability in dynamical representation learning
can come from the **structure of the latent code**.

Thank you. Questions welcome.

Backup slides

Backup: statistical protocol

- **Point estimates:** interquartile mean (IQM) over seeds within each system, then mean across systems.
- **Paired effect (MSE):** $\Delta = \log_{10}(E_{\text{cand}}/E_{\text{Dense}})$, one-sided alternative $\Delta < 0$.
- **Multibasin & support diagnostics:** unit = seed; within-system Wilcoxon; Holm across systems.
- **Dysts:** unit = system; exact one-sided sign test; a superscript means improvement on all 10/10.
- **Intervention panels:** descriptive mechanism diagnostics (bootstrap / IQR bands), not tests.

Backup: training details

Controlled multibasin

- rollout $L = 8$, 200k steps, batch 256, AdamW
- lr: enc/dec 5×10^{-5} , transition 5×10^{-6}
- $\lambda_{\text{pred}} = 1$, $\lambda_{\text{rec}} = 0.03$, $\lambda_{\text{lin}} = 1$, $\lambda_{\text{sp}} = 3 \times 10^{-3}$
- LISTA: threshold $\alpha = 0.15$, 2 refinement loops
- label-free boundary-emphasized resets

Dysts $dt \times 30$

- rollout $L = 10$, 100k steps, batch 256
- $\lambda_{\text{sp}} = 6 \times 10^{-3}$ (dense: 0)
- deterministic caches; disjoint train/val/test
- best re-encoding period $m \in \{10, \dots, 200\}$
- $d_z = 256$ latent dimension throughout